



INSTITUTO POLITÉCNICO NACIONAL
ESCUELA SUPERIOR DE CÓMPUTO



Introduction to Cryptography

Session 13: Schoolbook RSA

June 10, 2026

As usual, please use only one programming language: C/C++, Java or Python to develop the following programming exercises.

1. Random prime numbers and RSA key generation

Every student must do the following exercises

1. Use a cryptographic library for the programming language you have chosen to generate a prime number for each size listed below.
 - a) 16 bits
 - b) 32 bits
 - c) 512 bits
 - d) 2048 bits

Print a prime number for each size.

2. Design and implement a program to generate a key pair for schoolbook RSA. Do your own implementation, **do not use a function of the programming language that already generates a key pair for RSA**. Your function must fulfill the following requirements:
 - It must receive as parameter the size in bits of the prime numbers. This size could be 32, 512 or 2048 bits.
 - Both prime numbers must be of the same size i.e $|p| = |q|$, but they must be different i.e. $p \neq q$.
 - Choose at random the value of the public exponent e . DO NOT USE a value of $e = 3$ or $e = 65537$. Test that $\gcd(e, \phi(n)) = 1$. Use a gcd function of the programming language to test this.
 - Use a function included in the programming language to find the multiplicative inverse of e , i.e. $ed \bmod \phi(n) = 1$.

- Print the value of the public key (e, n)
- Print the value of the private exponent d .

2. Enciphering and deciphering using RSA

Design and implement the following programming exercises in pairs.

1. Use the function you implemented in Section 1, to generate a key pair for RSA. Copy the values of the public key and the private key in a file.
2. Design and implement a program to encipher a value of 16 bits (e.g. a key for TBC16) using RSA. **DO NOT use a function of the programming language that already enciphers using RSA.** Do the following steps:
 - a) Ask the user the public key, i.e. (e, n) .
 - b) Generate a random number r of 16 bits and print it as an integer. (Note, if we represent r as an integer $r \in [0 - 2^{16}]$)
 - c) Use the public key given by the user to encipher $r : c = r^e \bmod n$. Print the value of c as a integer
3. Design and implement a program to decipher a ciphertext c using RSA. Do your own implementation, **DO NOT use a function of the programming language that already deciphers using RSA.** Do the following steps:
 - a) Ask the user the private key, i.e. d and n .
 - b) Ask the user the value of c .
 - c) Use the private key given by the user to decipher $c : m = c^d \bmod n$. Print the value of d as an integer.
4. To test your functions you must do the following:
 - a) Each student must generate his/her own key pair for RSA. Then each student must share his/her public key with his/her partner.
 - b) Each student must separately encipher his/her own key for TBC with RSA and share the ciphertext c with his/her partner.
 - c) Each student must separately decipher the c given by his/her partner to recover the secret key for TBC.

3. Products

Each student must write his/her own report with the following information:

1. Your personal information, date of the lab session and the topic that we are studying in this lab session.
2. Include the source code of the functions you designed and implement for exercises of Section 2.
3. Include screenshots of your programs running.